Exercise 3.16

1. Graph the following quadratic equations and state their nature of solutions.
   (i) $x^2 - 9x + 20 = 0$ (ii) $x^2 - 4x + 4 = 0$ (iii) $x^2 + x + 7 = 0$
   (iv) $x^2 - 9 = 0$ (v) $x^2 - 6x + 9 = 0$ (vi) $(2x - 3)(x + 2) = 0$
2. Draw the graph of $y = x^2 - 4$ and hence solve $x^2 - x - 12 = 0$
3. Draw the graph of $y = x^2 + x$ and hence solve $x^2 + 1 = 0$
4. Draw the graph of $y = x^2 + 3x + 2$ and use it to solve $x^2 + 2x + 1 = 0$
5. Draw the graph of $y = x^2 + 3x - 4$ and hence use it to solve $x^2 + 3x - 4 = 0$
6. Draw the graph of $y = x^2 - 5x - 6$ and hence solve $x^2 - 5x - 14 = 0$
7. Draw the graph of $y = 2x^2 - 3x - 5$ and hence solve $2x^2 - 4x - 6 = 0$
8. Draw the graph of $y = (x - 1)(x + 3)$ and hence solve $x^2 - x - 6 = 0$

3.9 Matrices

Introduction

Let us consider the following information. Vanitha has 12 story books, 20 notebooks and 4 pencils. Radha has 27 story books, 17 notebooks and 6 pencils. Gokul has 7 story books, 11 notebooks and 4 pencils. Geetha has 10 story books, 12 notebooks and 5 pencils.

Details	Story Books	Note Books	Pencils
Vanitha	12	20	4
Radha	27	17	6
Gokul	7	11	4
Geetha	10	12	5

Now, we arrange this information in the tabular form as follows.

$$\begin{array}{l} \text{First row} \\ \text{Second row} \\ \text{Third row} \\ \text{Fourth row} \end{array} \begin{pmatrix} 12 & 20 & 4 \\ 27 & 17 & 6 \\ 7 & 11 & 4 \\ 10 & 12 & 5 \end{pmatrix}$$

First Column, Second Column, Third Column

Here, the items possessed by four people are aligned or positioned in a rectangular array containing four horizontal and three vertical arrangements. The horizontal arrangements are called "rows" and the vertical arrangements are called "columns". The whole rectangular arrangement is called a "Matrix". Generally, if we arrange things in a rectangular array, we call it as " Matrix".

Applications of matrices are found in several scientific fields. In Physics, matrices are applied in the calculations of battery power outputs, resistor conversion of electrical

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